30017 Corporate Finance
Hannes Wagner

The included PDF files are examples of case study write-ups made by students of the 30017 Corporate Finance course in 2012-2013. The underlying case was "Hutchison Whampoa Limited: The Capital Structure Decision". The write-ups were evaluated as "excellent" and the students have agreed for their work to be distributed. All rights to their work remain with them.

The instructions that students received were the following:

"Your assignment is to provide a written analysis of the case. We will discuss the case in class as indicated in the syllabus. All team members should be knowledgeable about the facts of the assigned case. While dealing with the decision problem at hand, your report should address the following questions:

1. Assume Hutchison Whampoa will require US$ 1 billion of financing in 1996. Also assume that new equity can be raised at $48.8 a share and that a long-term debt issue will carry an interest cost of HIBOR plus 70 basis points (bps). How would an equity or debt issue impact on Hutchison's financial position and performance?
2. Assess Hutchison Whampoa's current capital structure in light of its future financing needs.
3. What bond rating do you think Hutchison Whampoa will be able to obtain from S&P?
4. Compare the debt financing options. Explain why you are for/against the Yankee bond option.
5. What kind of capital structure would you propose to Hutchison Whampoa and why?"

Group name: $Wagner
Balzola, Guglielmo (1607498) – class 17; Borsos, Robin (1564970) – class 17; Jakab, Tamas (1579747) – class 18; Pallotta, Matteo (1562016) – class 16; Ragini, Raffaele (1570472) – class 18

HUTCHISON WHAMPOA CASE

GROUP REPORT

EXECUTIVE SUMMARY

After careful analysis, the finance department of Hutchison Whampoa Limited (HWL) compared two proposed financing options: USD 1 billion equity issuance and the equivalent debt issuance. In light of its current capital structure and the expected BBB rating, we propose that the company diversifies its capital structure. This should be done by increasing its leverage, issuing debt in the form of Eurobonds and Hong Kong bonds, while keeping its investment grade bond rating. If the company risks losing its rating, then more capital can be raised by equity issuance, but maintaining the control of the main shareholders.

SUMMARY OF FACTS

We have summarized the relevant details in Exhibit 1 in order to maintain transparency.

STATEMENT OF PROBLEM

In order to finance and maintain the forecasted impressive growth, HWL needs to raise external financing. The main question is that the required financing of USD 1 billion (for 1996) be raised by equity issuance or by long-term debt issuance. In order to choose the best option, we have to assess HWL's current and prospective capital structure, compare the available debt options (also predicting the rating agency's evaluation of HWL) and propose an adequate capital structure for the future.

ANALYSIS AND RECOMMENDATIONS

COMPARING THE DEBT AND EQUITY ISSUE OPTIONS

In order to compare the financing options, we always refer to Exhibit 2. The company has 3610 million shares outstanding[1]. HWL needs USD 1 billion, which is equal to HKD 7.76 billion. A successful evaluation of the financing options requires forecasts for the year 1996. Considering the growth opportunities, and the past years, we forecasted a conservative profit growth rate of 12%.

EQUITY ISSUANCE

The company issues shares at the price of HKD 48.8. This means that the company issues 159.02 million shares, thus increasing the total share capital to HKD 66 599 million. Assuming that the P/E ratio is constant, EPS increases to 2.84. This increase is due to the fact that the positive effect of increasing profitability outweighs the negative effect of new issuance. The profitability and payout rate increase also prevails in case of dividends, so dividends per share increases to 1.34. Secondly, we predict that the stock price moves in an interval specified in Exhibit 2 with expected price of HKD 43.49[2]. However, since the company has more equity, the issuance influences ROE in a negative way.

DEBT ISSUANCE

The debt issuance enjoys the benefit of tax shield. In fact, given our profit forecast, the effect influences the financials in the following way. The profit is lower in nominal terms[3], the P/E ratio is higher and the expected stock price is also higher, HKD 45.97[4]. Since there is no new equity issuance, both EPS and Dividends per share are higher than in the case of equity issuance.

---

[1] We calculated the shares outstanding as Earnings/EPS, this is consistent with the data given for the last years.

[2] We computed the stock price the following way: the expected stock price is EPS*P/E ratio, and used the last years' intervals to predict highs and lows. We calculated this with the assumption of profit growth equal to 12%, and thus the expected stock price is lower than the one at which the company issued shares. The expected price of the shares would be equal to 48.8 with 25% profit growth which we consider as unreasonable.

[3] The profit will be lower by the amount of tax shield.

[4] As Footnote 2.

Undoubtedly, ROE, in compliance with the risk for equity holders, increases due to the increased leverage.

HUTCHISON WHAMPOA'S CAPITAL STRUCTURE AND FUTURE FINANCING NEEDS

HWL has capital commitments for 1995 equal to HKD 30205.1 million. Since the company does not have enough internally generated funds for the total amount, it has to raise additional financing. Due to the nature of its business, the earnings of the company are stable and expected to increase. In addition, the company is only slightly indebted. Therefore, the opportunity of higher leverage could be exploited. Also, increased debt lowers the WACC of the company, thus increasing ROE. Nevertheless, the company has solid financial situation and therefore it would be relatively easy for it to find debt financing at reasonable cost. An increased amount of debt and commitments due to it will prevent management from possibly harmful overinvestment. This could in turn increase operating efficiency. Another concern of the shareholders is control; thus by issuing more shares, the main shareholders should retain their influence.

HWL'S EXPECTED BOND RATING

The predicted rating is based on a well-established method by S&P. According to our analysis, HWL will receive a rating of BBB from the agencies, which means that they consider its bond as an investment-grade corporate bond. The details can be found in Exhibit 3.

COMPARING THE DEBT FINANCING OPTIONS

All things considered, Yankee bonds are an attractive option. They have an advantage considering the size, scope and influence of the US Bond market. Also, they are part of an increasingly growing liquid market. Nevertheless, compliance with tough SEC requirements can only be avoided if the bond was offered for a restricted pool of investors. Thus, Eurobonds have a cost advantage over Yankee bonds.

The firm is now evaluating different debt financing options, summarized in the following table:

Comparing debt financing options			
Type of financing	**Maturity**	**Pricing**	**Comments**
Syndicated loan	5-7 years	Volatile	• Feasible option • Shorter maturity • Relatively volatile HIBOR increases unpredictability
Hong Kong bonds	≥ 10 years	High	• HKD denomination, no exchange rate risk • HWL is a recognized name on the market • No additional costs • High financing needs of Chinese firms have an upper pressure on pricing
Eurobonds	≥ 10 years	Good	• Broad investor base • Diversification • Tax advantages • High compliance and advertising costs • Less rigorous requirements than in the case of SEC in the US
Yankee Bonds	≥ 10 years	Good	• Similar features as the Eurobond • Considerable SEC requirements (although can be avoided if sold for a selected investor base)

THE PROPOSED CAPITAL STRUCTURE

The company has clearly reached the limits of funding growth mainly by internally generated funds. Thus, in order to maintain high growth rate, HWL has to attract external capital. In the following years, the company should evaluate how flexible it is financially. This means that the company has to maintain its credit rating while benefiting from a more diversified capital structure. In particular, both historical stability of earnings and future growth implies that raising more debt would not result in safety issues. Therefore, HWL should raise as much debt as possible while maintaining its BBB rating, because otherwise the bond would become a junk bond and the interest rate increase considerably[5]. In case this amount is not enough, equity issuance is required, while not losing control of the main shareholders. What regards the debt; we suggest issuance of both Eurobonds and Hong Kong bonds. Eurobonds have tax advantages and offer the advantage of currency diversification, and Hong Kong bonds offer the ease of issuance and sale.

Nevertheless, if there was no risk of losing its investment grade status, HWL should finance its growth from debt. With our proposal, the company can exploit diversified capital structure.

[5] According to a J. P. Morgan analysis, the spread between U. S. investment grade bonds and junk bonds can be as high as 2.2% (source: http://www.jpmorganinstitutional.com/cm/)

Exhibit 1

Required financing (USD mln)	1000
Required financing (HKD mln)	7760
With yearly avg. exchange rate of	7,76
Corporate tax rate (source: KPMG)	16,5%

	1995	1994
Shares outstanding (= Earnings/EPS)	3610,19	3613,06

Forecasts

Profit growth

Year	Profit attributable to the shareholders (HKD mln)	Change %
1996 (forecast)	10715	12,00%
1995	9567	19,27%
1994	8021	

Dividend growth

Year	Payout rate	Change %
1996 (forecast)	47,10%	2,50%
1995	44,60%	2,69%
1994	41,91%	2,88%

Proposals

Equity issuance		**Debt issuance**	
The company issues new shares at (HKD)	48,80	3-month HIBOR (June 1996)	5,32%
Number of shares issued (mln)	159,02	Spread	0,70%
Total capital raised (HKD mln)	7760,00	Interest rate	6,02%
Total shares outstanding (mln)	3769,21		
Total share capital equals (HKD mln)	66599,00	Total capital raised (mln HKD)	7760

Exhibit 2			
	1995	1996 Equity	1996 Debt
Profit Attributable to the shareholders	9567	10715	10638
Dividends	4267	5047	5011
Profit for the Year Retained	5300	5668	5627
EPS	2,65	2,84	2,95
Stock price (High)	47,10	48,18	50,65
Stock price (Expected)	40,55	43,49	45,97
Stock price (Low)	34,10	38,81	41,28
P/E Ratio	15,3	15,3	15,6
Dividends per share	1,18	1,34	1,39
ROE	16,50%	16,09%	18,08%

Exhibit 3								
	U.S. Industrial long-term debt						HWL	
Three year (1994 to 1996) medians	AAA	AA	A	BBB	BB	B	1995 ratios	rating
EBIT interest coverage (x)	16,10	11,10	6,30	4,10	2,30	1,20	3,7	BBB
EBITDA interest coverage (x)	20,30	14,90	8,50	6,00	3,60	2,30	4,9	BBB
Funds from operations/total debt (%)	116,40	72,30	47,50	34,70	18,40	10,90	14,8	BB
Free operating cash flow/total debt (%)	76,80	30,50	18,80	8,40	2,40	1,20	7,3	BBB
Pretax return on capital (%)	31,50	23,60	19,50	15,10	11,90	9,10	12,4	BB
Operating income/sales (%)	24,00	19,20	16,10	15,40	15,10	12,60	17,8	AA
Long-term debt/capital (%)	13,40	21,90	32,70	43,40	53,90	65,90	28,9	A
Total debt/capitalization (%)	23,60	29,70	38,70	46,80	55,80	68,90	44,8	BBB

BIEMF – Class 17

Hutchison Whampoa Limited - Capital structure decision

From: Benaglio Benedetta, Gattone Tulia, Leonardi Edoardo, Macaccaro Lucia, Savino Martina

To: CFO of Hutchison Whampoa LTD.

Executive Summary

The aim of this report is to provide advice regarding the long-term financing strategy and its effects on the overall capital structure of Hutchison Whampoa Ltd. The estimated quantity required for future investments, which is around HK$39 billion[i], is expected to be repaid in no less than ten years, thus the syndicated bank loan option is not feasible, both in terms of amount and timespan. For these reasons, we would recommend an issuance, in the form of Eurobonds, of the maximum amount of debt allowing to obtain an investment-grade rating of BBB. The residual should be raised through equity.

Summary of facts

As customary to most of large Hong Kong firms, in the past HWL has strongly relied on the plowback of its internally generated earnings to foster growth - see the relevance of shareholders' equity in the capital structure of the firm in Exhibit 1. Up until now, this has been sufficient to satisfy its financing needs, due to the fact that its previous projects were characterized by short and medium-term maturities and revenues could be forecasted over this timeframe. Yet, the company is currently planning to implement long-term ventures; hence, such a capital structure is no longer appropriate. Moreover, some of the already contracted capital commitments, such as investments in telecommunications and infrastructures, are likely to start providing revenues in more than a year from now, presenting thus negative cash flows in the early years of their realization. This requires enough funding to be sustainable.

Statement of problem

HWL is faced with the choice of raising additional capital either by issuing new shares or underwriting additional debt, which could be in the form of straight debt, Eurobonds or Yankee bonds.

Analysis

We have proceeded with the analysis of both possibilities in order to raise HK$7.8 billion.

- Scenario 1: Equity issuance

By deciding to resort to all-equity financing, given the share price of HK$48.80, the company would need to deliver 160,000,000[ii] new shares. Considering the pre-issuance number of shares outstanding - 3,616,101,695[iii] -, this would cause a fragmentation of the ownership of the company quantifiable around 4.24%[iv]. Furthermore, there would be also a dilution of earnings per share, partially offset by the increase in the value of the company provided by the newly exploitable long-term investments. This rise in shareholders' equity - HK$7.8 billion – is more than enough to cover the shortfall from the funding of the contracted projects – HK$6,468.9 million. Nevertheless, the company will be required to increase the overall amount of dividends, since it is not advisable to dilute them for the new number of shares, as this would imply a reduction in dividends per share in addition to the already lower EPS (see Table 1), which might provoke a negative reaction in the expectations of shareholders.

- Scenario 2: Debt issuance

The issuance of debt at an interest rate of 6.02%[v] would present interesting advantages for the business, since it would allow the company to benefit from a tax shield of HK$77.477,400[vi] and could present some flexibility in the choice of the form of debt to be used, as mentioned above. Furthermore, this would also determine better figures for the most important ratios – EPS, ROE and after-tax WACC - with respect to the previously analyzed scenario (see Table 1). Delivering straight debt would represent

a limitation to the firm in the sense of the market scope it is aiming at and the fierce competition from the high-yield paper introduced from the neighbouring PRC, so we do not advise to operate in this way. We surely don't believe Yankee bonds, whose market seems to have been in free-fall in the last year (see Exhibit 2), would be a proper solution either. In fact, this is probably the most difficult kind of bond to issue, given the necessity of filing a registration statement with the SEC and of disclosing the company's financial statements in compliance with the US GAAP, which would be time-consuming and represent additional costs. In fact, we do not believe institutional investors would be willing to have SEC rule 144A applied, given the low international recognition of the firm. Moreover, the company would be exposed to a serious currency risk, due to the different denomination. On the other hand, we believe that it would be of utmost convenience for the firm to issue debt in the format of Eurobonds, which can be sold internationally with a HK$ denomination, presenting a large and expanding market - we believe it shall recover from the slight decline of last year - and for which no additional costs are needed. Despite this aspect, raising additional debt presents the great disadvantage of increasing the financial risk borne by the company's shareholders. Shareholders might even feel that this risk exceeds the loss in ownership share which would derive from the equity issuance. This is a serious argument to be considered, given the precarious situation of HWL rating. Starting from our comparative analysis presented in Table 2, we think the company would now receive a BBB rating, therefore classifying as investment grade. Nevertheless, the increase in debt might trigger a downgrade, due to the additional risk, towards the BB level, which represents the first step of the infamous junk bond level. See Table 3 for a sum up of the main characteristic of the here presented kinds of debt.

Recommendations

We suggest the company to take up as much debt as it can with respect to the downgrading argument presented above, to exploit the tax advantage and avoid the dispersion of ownership, and finance the rest of the funding with the issuance of equity, unlikely to affect the market perception of the firm.

Exhibits, Tables and Notes

Exhibit 1 - Capital structure of the firm (Capital employed = 100%) in 1995

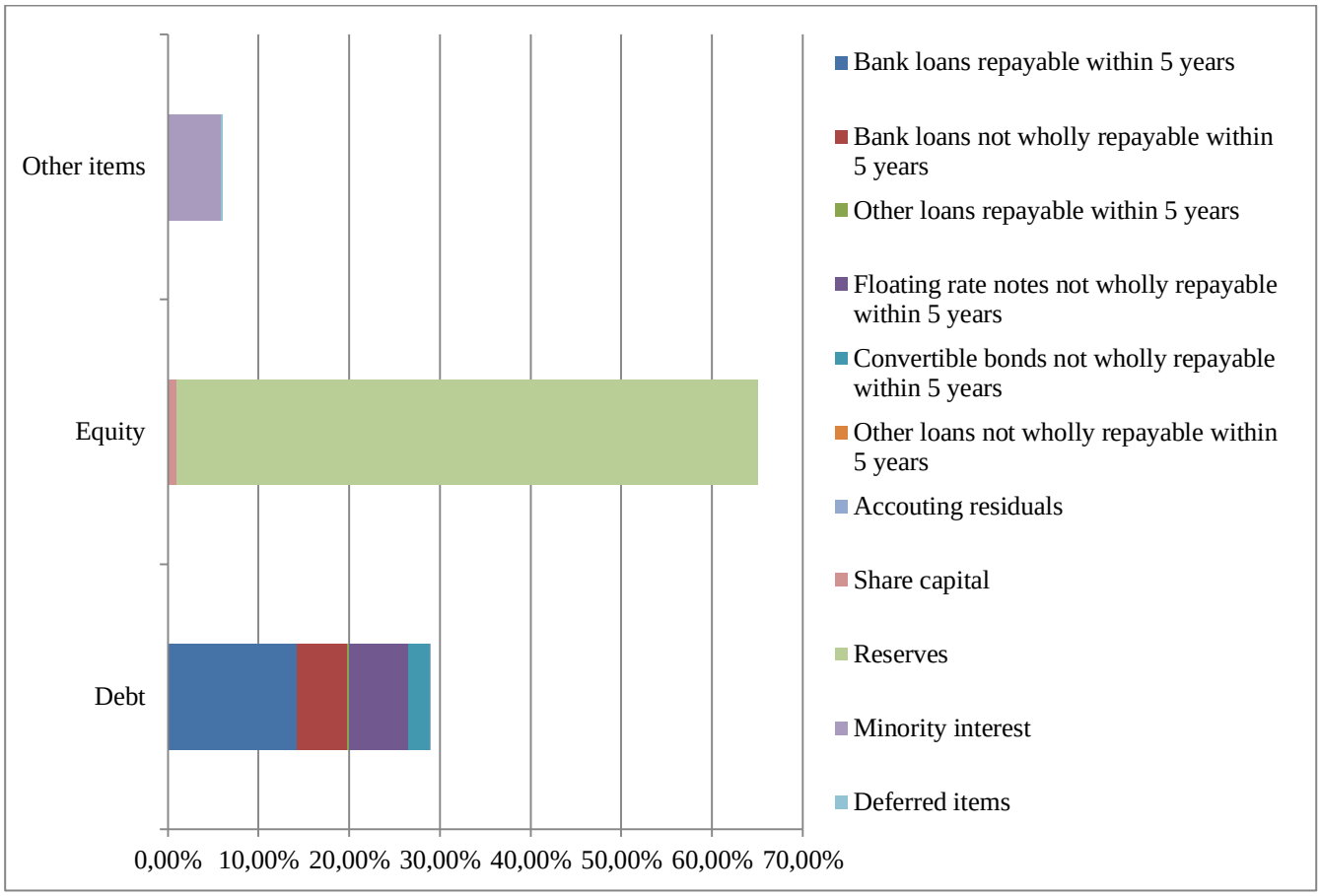


Table 1 – Ratios comparison between the two scenarios[vii viii]

	All equity financed	All debt financed
EPS	2.53[ix]	2.54[x]
EBIT interest coverage	3.98[xi]	3.41[xii]
Pretax return on capital (%)	8.52[xiii]	8.04[xiv]
Long term debt/Capital (%)	26.62[xv]	34.55[xvi]
ROE (%)	14.36[xvii]	15.59[xviii]
Dividends per share[xix]	1.18	1.18
After-tax WACC (%)	11.08[xx]	11.07[xxi]

All highlighted values represent the better figures between the two scenarios

Table 2 - Comparative analysis of competitive firms and average ratios for the various rating classes

	Hutchison Whampoa	Swire Pacific Ltd.	Wharf Holdings Ltd.	Jardine Matheson
EBITDA interest coverage	4.9	5.5	3.8	6
EBIT interest coverage	3.7	3.8	2.5	3
Funds from operations/Total debt (%)	14.8	31.7	63	35.7
Free operating cash flow/Total debt (%)	7.3	30.3	79.6	67.9
Pretax return on capital (%)	12.4	8.6	4.3	11.3
Operating income/Sales (%)	17.8	13.5	NA	3
Long term debt/Capital (%)	28.9	26.4	21.7	12.4
Total debt/Capitalization (%)	44.8	32	22.6	24.8

Legend
AAA
AA
A
BBB
BB
B
Unrated

Exhibit 2 - Bond Market Development

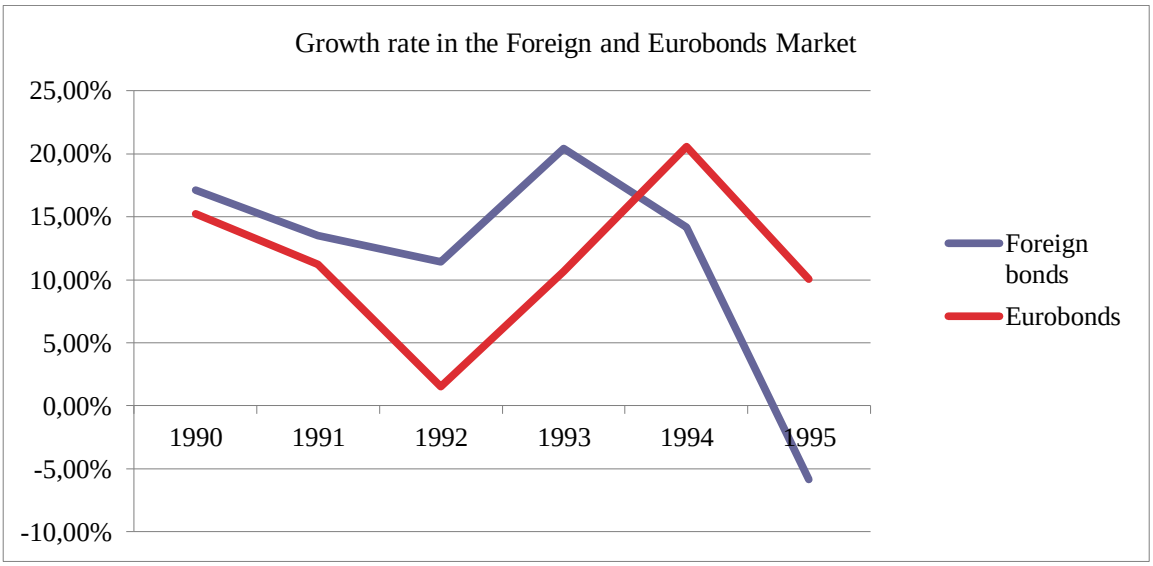


Table 3 - Bond comparison matrix

Bond type/Bond characteristic	Market scope	Risk	Accessibility	International recognition
Straight debt	Honk Kong market, flooded with cheap paper from PCR	Liquidity risk	Easily accessible, wide firm recognition at the domestic level	Underwritten only by domestic banks
Eurobonds	International, both for the primary and the secondary market	In slight decline in the last year, but expected to take up again	Easily accessible, no need for additional disclosure or costs of compliance	Underwritten by an international syndicate of banks
Yankee bonds	International in the primary market, domestic capital markets for the secondary one	Currency risk, slumping market	High entry costs, need for registration with the SEC and compliance with the American GAAP	Underwritten by a syndicate of banks of the US

Notes:

[i] HK$39 billion=US$5 billion

[ii] Required Founding / Share Price = US$1bl / (HK$48.80 / HK$/US$7.8$) = 160,000,000

[iii] Dividends / Dividends per share = 4,267ml / 1.18 = 3,616,101,695, values are taken from the 1995 Consolidated Profit and Loss Account

[iv] Number of Shares / Total number of shares outstanding = 160,000,000 / (3,616,101,695 + 160,000,000) = 4.24%

[v] Interest Rate = (HIBOR+ 70bps) = 5.32% + 0.7% = 6.02%

[vi] Tax Shield = Tax Rate * Debt * Interest Payment = 16.5% * HK$7.8bn * 6.02% = HK$77,477,400, assuming a corporate tax rate of 16.5% as in "Sherwood,SG 1997, "Tazing decision", *China Business Review*, 24, 5, p.42, Business Source Complete, EBSCOhost, viewed 15 May 2013

[vii] All calculations are based on the assumption that the share price is equal to HK$48.8

[viii] We opted for taking into account only data referring to the Balance Sheet of 1995 instead of forecasting the one of 1996, since given the recent expansion of the company, the likelihood of being inaccurate would have been extremely high

[ix] EPS = (Profit attributable to the shareholders)/(Number of shares outstanding)=(HK$9567ml/3,616,101,695+160,000,000)

[x] EPS = (Profit attributable to the shareholders – Interest Expense Net of Tax Shield)/(Number of shares outstanding) = (HK$9,567ml + HK$7,800ml*0.0602*(1-0.165))/3,616,101,695

[xi] EBIT Interest Coverage = (EBIT/Interest Payment) = (HK$11,181ml/HK$2,808ml)

[xii] EBIT Interest Coverage = [EBIT/(Interest Payment + Interest Payment on new debt)] = [HK$11,181ml/(HK$2,808ml+HK$469.56ml)]

[xiii] Pretax Return on Capital = [(EBIT-Interest Payment)/ Capital]% = [(HK$11,181ml-HK$2,808ml/(HK$90,458ml+HK$7,800ml)]%

[xiv] Pretax Return on Capital = [(EBIT- Interest Payment – Interest Payment on new debt)/(Capital)]% = [(HK$11,181ml – HK$2,808ml - HK$7,800ml*0.0602)/(HK$90,458ml+HK$7800)]%

[xv] Long-term Debt/Capital = [HK$26153ml/(HK$90458ml+HK$7800ml)]%

[xvi] Long-term Debt/Capital = [(HK$26,153ml+HK$7,800ml)/(HK$90,458ml+HK$7,800ml)]%

[xvii] ROE = [(Profit attributable to shareholders)/(Shareholders' funds)]% = [HK$9,567ml/(HK$58,839+HK$7,800ml)]%

[xviii] ROE = [(Profit attributable to shareholders - Interest Expense Net of Tax Shield)/(Shareholders' funds)]% = [(HK$9,567ml - HK$7,800ml*0.0602*(1-0.165))/HK$58,839]%

[xix] This value is assumed to be constant with respect to 1995 since a change in the payout policy would not be advisable due to the negative response of shareholders

[xx] After-tax WACC = [(1-Tax Rate)*Interest Rate*(Debt/Total Capital) + ROE*(Equity/Total Capital)]% = (1-0.165)*0.0602*[HK$26,153ml/(HK$90,458ml+HK$7,800ml)]+0.1436*[(HK$58,839ml+HK$7,800ml)/(HK$90,458ml+HK$7,800ml)]

[xxi] After-tax WACC = [(1-Tax Rate)*Interest Rate*(Debt/Total Capital) + ROE*(Equity/Total Capital)]% = (1-0.165)*0.0602*[(HK$26,153ml+HK$7,800ml)/(HK$90,458ml+HK$7,800ml)]+0.1436*[HK$58,839ml/(HK$90,458ml+HK$7,800ml)]

HUTCHISON AND WHAMPOA CASE

GROUP NAME: STANDARD OIL

BY ADRIAN BERGEM, LUCA A. RIIS, SJUR RESALAND AND CHRISTIAN D. SOLBERG

NORWEGIAN SCHOOL OF ECONOMICS ERASMUS EXCHANGE - CLASS 16/17

EXECUTIVE SUMMARY

Hutchison Whampoa needs additional sources of financing in order to sustain its current growth rate. As not to dilute the controlling shares of Cheung Kong Holdings too significantly, the issue of new stocks should be no more than approximately 40% of the total financing needed. This percentage of equity-issuing is further supported by our forecasting model of maximizing EPS to the shareholders - the interest costs are not sufficiently offset by the tax shield created by the debt as long as the tax rate does not exceed 16%. On the other hand, according to the Pecking Order Theory, debt should be issued before equity in order to send the right signals to the market. Thus a majority of 60% of the financing needs should be met through the issue of bonds. The forecasted bond rating of A- will most likely give the company access to highly liquid international bond markets. It is recommended that Hutchison Whampoa issues Eurobonds (HKD) due to its advantages of freedom of regulations and less strict disclosure requirements, amongst others.

SUMMARY OF FACTS

Different aspects and development in the markets of the current capital structure impel the firm to seek alternatives. Normally, the issuing of new equity has been avoided in order not to endanger current shareholders' position. Furthermore, banks have had a policy of limiting their exposure to single firms and have only offered fluctuating interest rates. The straight debt home market is highly saturated with high yield-paper. A potential bond-issuing in the international bond market will give

access to a highly liquid market with many benefits such as fixed interest rates and it will widen the firm's recognition worldwide.

STATEMENT OF PROBLEM

Hutchison and Whampoa has mainly financed its business with retained earnings (approximately 40% payout ratio for the last 3 years) and short- and medium term bank financing (total debt/capital-ratio = 44,8% as opposed to long-term debt/capital = 28,9%). As the firm is growing and is taking on longer-term projects , it will need a new and different source of funding.

ANALYSIS

Exhibit 1 shows a comparison of a 1 billion USD financed either with an equity-issuing or a debt-issuing. It is assumed that the information about the issuing is accounted for in the current stock-price, i.e. the market is at least weakly efficient. The model looks at the EPS and ROE between the two scenarios ceteris paribus. The EPS-ratios diverge as the tax-rate increases and hence the tax-shield increases. As the tax-rate sensitivity analysis shows, the break-even point comes at a rate of 15,96%. On the other hand, ROE obviously favors debt issuing. Furthermore, the present value of bankruptcy costs is not included and would have worked against debt-issuing, although, with a current very low leverage, the marginal costs of bankruptcy costs would have been initially less significant.

The cost of debt is highly dependant on the credit rating Hutchison and Whampoa will be given. Table 1 shows some solidity-measures for the firm measured up against comparable companies and their according rating based on S&P long-term debt rating. Each corrisponding measure for Hutchison and Whampoa is then weighted in the same way that has been done to the other firms. An overall rating of A- seems plausible. With such a strong credit rating, a potential issuing in the international bond market will be of interest.

Table 2 shows different debt instruments and their features. The preference of a fixed interest-rate and access to a highly liquid market, the international bond market seems to be the only option for

debt issuing. The trading of USD creates a further exchange risk that could be avoided using Eurobonds with HKD, though it also entails a diversification of currency and thus a potential decrease in systematic risk. Finally, the process of issuing publicly-traded foreign bonds is demanding - it can take over 3 months.

Hutchison Whampoa has currently a debt to capital ratio of 28.9%. High debt to capital ratios may constrain the companies flexibility and increase the risk of default. Exhibit 3 presents different financing alternatives and their impact on both the debt to capital ratio and the control position of Cheung Kong Holdings. The trade-off between the dilution of CKH´s controlling shares and the leverage level must be taken into account when deciding on a certain capital structure. It should not stray too far away from the leverage level of comparable firms. If our forecasts regarding Hutchison Whampoa's credit rating is to hold, then the long-term debt/capital ratio should not surpass 43.4% (BBB level)

The pecking order theory states that the firm should finance itself firstly internally, then through debt and finally through equity in terms of sending the right signals to its shareholders. This will certainly also have a great importance when deciding the capital structure.

RECOMMENDATIONS

Based on the analysis it is recommended that Hutchison Whampoa obtains a capital structure comprising of approximately 60% debt and 40% equity. Supported by a relatively favorable bond rating and having many advantageous features, a bond issue in the Eurobond market is proposed.

EXHIBITS AND TABLES ***EXHIBIT 1***

IN-DATA		
Shareholders' equity	HKD	58 839 000 000
Share price	HKD	48,80
Dividends	HKD	4 267 000 000
Capital needed	HKD	7 800 000 000
Net profits	HKD	10 048 000 000
EPS before issue		2,65
HIBOR		0,0532
Bps over HIBOR		70
Tax rate (1)		10,13 %

1) Tax-rate set to the same as 1995, calculated as taxation as a % of profits before taxes
2) (Net profits - Divindends) / EPS before issue
3) Capital needed / share price
4) HIBOR + (Bps over HIBOR/10000)
5) Capital needed * effective interest rate
6) Total interest * tax rate
7) Total interest * (1-tax rate) / shares outstanding
8) (Net profits - dividends) / (Shares outstanding + new stock issue)
9) (Net profits - dividends) / (Shares outstanding + new stock issue)
10) Share price / EPS
11) Net profits / (Shareholders' equity + new stock issue*share price)
12) Net profits / Shareholders' equity

CALCULATIONS

Equity issuing		**Debt issuing**		
Shares outstanding (2)	2181509434	**Effective interest rate (4)**		0,0602
New stock issue (3)	159836066	**Total interest (5)**	HKD	469 560 000
% of new issue over shares outstanding	7,33 %	**Tax shield (6)**	HKD	47 566 428
		After-tax interest per share (7)		0,19344

RESULTS

EPS after issue of equity (8)	2,4691	**EPS after issue of debt (9)**	2,4566
P/E after issue of equity (10)	19,7643	**P/E after issue of debt (10)**	19,8652
ROE after issue of equity (11)	15,08 %	**ROE (12)**	17,08 %

EXHIBIT 2

Sensitivity analysis for the tax rate

Tax rate	EPS after issue of debt
10,00 %	2,4563
11,00 %	2,4584
12,00 %	2,4606
13,00 %	2,4627
14,00 %	2,4649
15,00 %	2,4670
16,00 %	2,4692
17,00 %	2,4713
18,00 %	2,4735
19,00 %	2,4757
20,00 %	2,4778

EXHIBIT 3

HKD 39 billions financed through (in HKD millions):

	Current	100% debt	80-20 debt-equity	60-40 debt-equity	40-60 debt equity	20-80 debt-equity	100% equity
Long term liabilities	26174	65174	57374	49574	41774	33974	26174
Capital	90458	129458	129458	129458	129458	129458	129458
Long term debt to capital ratio	28.93%	50.34%	44.32%	38.29%	32.27%	26.24%	20.22%
Cheung Kong Holdings per cent of Hutchison Whampoa	49.9%	49.90%	44.06%	39.44%	35.70%	32.61%	30.01%

Debt to capital ratio - comparables	
Swire	26.40%
Wharf Holdings	21.70%
Jardine Matheson	12.40%

TABLE 1

Rating	H&W				Swire Pacific Ltd.				Wharf Holdings Ldt.			
	1995	**1994**	**avg**	**rating**	**1995**	**1994**	**avg**	**rating**	**1995**	**1994**	**avg**	**rating**
EBIT interest coverage	3,7	4,5	4,1	BBB	3,8	3,6	3,7	BBB-	2,5	3,9	3,2	BB/BBB
EBITDA interst coverage	4,9	7,4	6,15	BBB+	5,5	5,4	5,45	BBB	3,8	5,1	4,45	BB/BBB
Funds from operations/total debt %	14,8	50,8	32,8	BBB-	31,7	27	29,35	BBB-	63	-32,7	15,15	BB-
Free operating CF/total debt %	7,3	11,4	9,35	BBB	30,3	4,6	17,45	A	79,6	43,8	61,7	AAA-
Pretax return on capital %	12,4	11,6	12	BB	8,6	7,6	8,1	B-	4,3	3,4	3,85	-
Operating income/sales %	17,8	14,9	16,35	A	13,5	11,2	12,35	B-				
Long-term debt/capital	28,9	29,5	29,2	A+	26,4	26,5	26,45	AA-	21,7	21,2	21,45	AA
Total debt/capitalization %	44,8	36,8	40,8	A-	32	31,8	31,9	AA	22,6	23,5	23,05	AA-

TABLE 2

Security/feature	Interest rates	Duration	Market for investors	Registration / Accounting policies	Currency
Straight debt	Fluctuating, increasing upward pressure on pricing	short and mid term	knowledgeable investor base, but becoming saturated	-	HKD
Eurobonds	Fixed	2-3 to 15 years or more	global, highly liquid, widening the firm's recognition to an international level	unregistered	any
Yankee bond (publicly issued)	same as similar-rated Eurobonds	any	global, highly liquid, widening the firm's recognition to an international level	Compulsory registration with the SEC and followingly the use of US GAAP (high accounting costs)	USD
Yankee bond (privately issued)	may be higher than similar-rated Eurobonds	usually 7 to 15 years or more	limited number of sophisticated investors thus less liquid	unregistered	USD

The Oracles of Glengyle

Luca Brogin	(class 16)
Max Sebastian Dovì	(class 16)
Leonardo Magnani	(class 17)
Edoardo Pricolo	(class 18)
Guglielmo Ventura	(class 16)

"HUTCHISON WHAMPOA LIMITED: THE CAPITAL STRUCTURE DECISION"

Case Study

Hutchison Whampoa Limited

Executive Summary

In order to face its future projects and maintain its present growth rate, Hutchison Whampoa will need approximately HK$ 40 bn over the next 5 years.

For this reason, the firm needs to adjust its future capital structure and, according to our analysis, it should finance its capital needs with debt issue, which will have a positive impact on the shareholders' position and on the firm's performance because the key ratios (EPS, ROE, DPS) will improve substantially and will enable Hutchison Whampoa to outdistance its competitors without compromising financial stability.

Furthermore, we have forecasted that the firm´s L-T corporate bonds will obtain from S&P's a BBB rating according to the parameters employed by the credit rating agency.

Finally, after having analysed all the debt options, we think that issuing Eurobonds is the most advantageous one: the market is deeper and broader than the Yankee bonds' one.

Summary of Facts and Statement of Problem

Hutchinson Whampoa Ltd is a publicly listed company based in Hong Kong and operating in a broad range of sectors. The major stake in the company is represented by Cheung Kong Holdings (49.9%). Historically Hutchinson has showed sustained profitability growth, providing valuable profits to its shareholders and a reliable flow of internal funds re-employed in investments.

The group's financial board has to assess whether the company's financial practices and current capital structure are capable of addressing the ongoing financial needs and of keeping the group on its virtuous growth path. In particular, if new external resources were needed, the decision would be between equity and debt financing and in the latter case it should be decided what type of debt instrument would best suit the company financial strategy.

Analysis & Recommendations

Hutchinson Whampoa is assumed to require HK$ 7.8 bn of financing for the year 1996; we analyze two different scenarios with their respective conditions:

How would an equity issue at HK$ 48.8 per share impact Hutchinson's financial position and performance?

Sensitivity analysis suggests that the equity issue would cause an increase in shareholders' funds to HK$66,639 millions along with an increase in the number of outstanding shares. Some of the resulting key ratios are show in **exhibit 1a.**

As far as long-term debt is concerned, the company would be able to borrow HK$ 7.8 bn carrying an interest rate of HIBOR (5.32%) plus 70 basis points. Thus, according to our estimate, the additional after-tax interest payment would be HK$392 million (**exhibit 1a** as well).

Comparison of the key ratios under the two scenarios shows that debt financing provides a better outlook.

The current capital structure of the firm can be easily inferred from **exhibit 2a**: Hutchison Whampoa is mainly financed by equity, with long-term debt being about half of the former.

We assumed a worst-case scenario i.e. a concave function describing the distribution of next-five-year capital commitments of about HK$ 40bn; by subtracting the values of retained earnings projections, we end up with the Net Financial Position function: there will be a need for external financing in years 1996-1998 for a total aggregate sum of approximately HK$ 7.5 bn (**exhibit 2b)**.

We can finally estimate the impact that the new stock of debt will have on profits, future retained earnings, shareholders' funds and so capital structure, as described by the Long-Term Debt to Capital ratio shown in **exhibit 2c**. From this forecast it is evident that no deterioration of debt sustainability can be expected, even because Hutchison is a profitable firm with a high degree of tangible assets and thus a lower cost of financial distress (Trade-Off Theory).

Key ratios, as illustrated in **exhibit 3,** provide the necessary guidance for a preliminary estimation of the likely rating that Hutchison Whampoa could obtain by Standard & Poor's.

Additional insight about the award of the likely rating can be gained if Hutchison Whampoa Ltd's ratios are examined jointly with comparable multi-sector diversified Hong Kong corporations of similar size, which have already been assigned a S&P's rating, namely Swire Pacific Ltd and Wharf Holdings Ltd. Hutchison Whampoa Ltd outperforms the other two corporations in profitability ratios and is practically equivalent to them in the interest coverage, but it lags seriously behind in the ratios related to debt exposure and therefore a lower rating should be expected.

Considering all these issues <u>we can expect Hutchinson Whampoa to be awarded a BBB rating</u>.

The Company's debt financing options are basically three, i.e. syndicated bank financing, straight debt and international bond market. Since banks usually lend at floating interest rates for maturities of 5 to 7 year, the first option doesn't suit Hutchison Whampoa's aims to have a fixed-rate, 10-year loan.

Furthermore the Hong Kong straight debt market is floated by more attractive high-yielding debt from PRC; the company is therefore left with the option of issuing securities in the fast growing international bond market.

Even if Yankee bonds and Eurobonds are to offer comparable yields, Hutchison should opt for an issuance of Eurobonds with a maturity of 10 years or more denominated in UK£ (to better match cash inflows and outflows from operations and offer currency diversification to investors), because a greater volume outstanding in nominal terms makes the Eurobond market more liquid, broad and deep than the Yankee bond market, and also because a lower bureaucratic burden and an increased speed of issuing Eurobonds can be expected. **Exhibits 4a** and **4b** show the volumes of the foreign bond and the Eurobond markets and their rate of growth worldwide.

Methodology and core assumptions:

I. The assumed rate of growth *g* is the one which equates NI_{1995} to NI_{1991}*(1+g)^4, i.e. ≈ 22%.
Profits attributable to shareholders (t)=4340*(1.218488714)^(t-1991).
II. The assumed payout ratio is the average of 1991-1995 payout ratios, i.e. ≈ 47%.
III. The tax rate is obtained by the KPMG historical chart of marginal tax rates paid by Hong Kong corporations.
IV. Cost of future long-term debt is assumed to follow the average effective one paid by the firm (≈ 11%).
V. Outstanding long-term debt is assumed to be repaid in constant installments every year.

Exhibit 1a: Key metrics in the debt scenario and equity scenario.

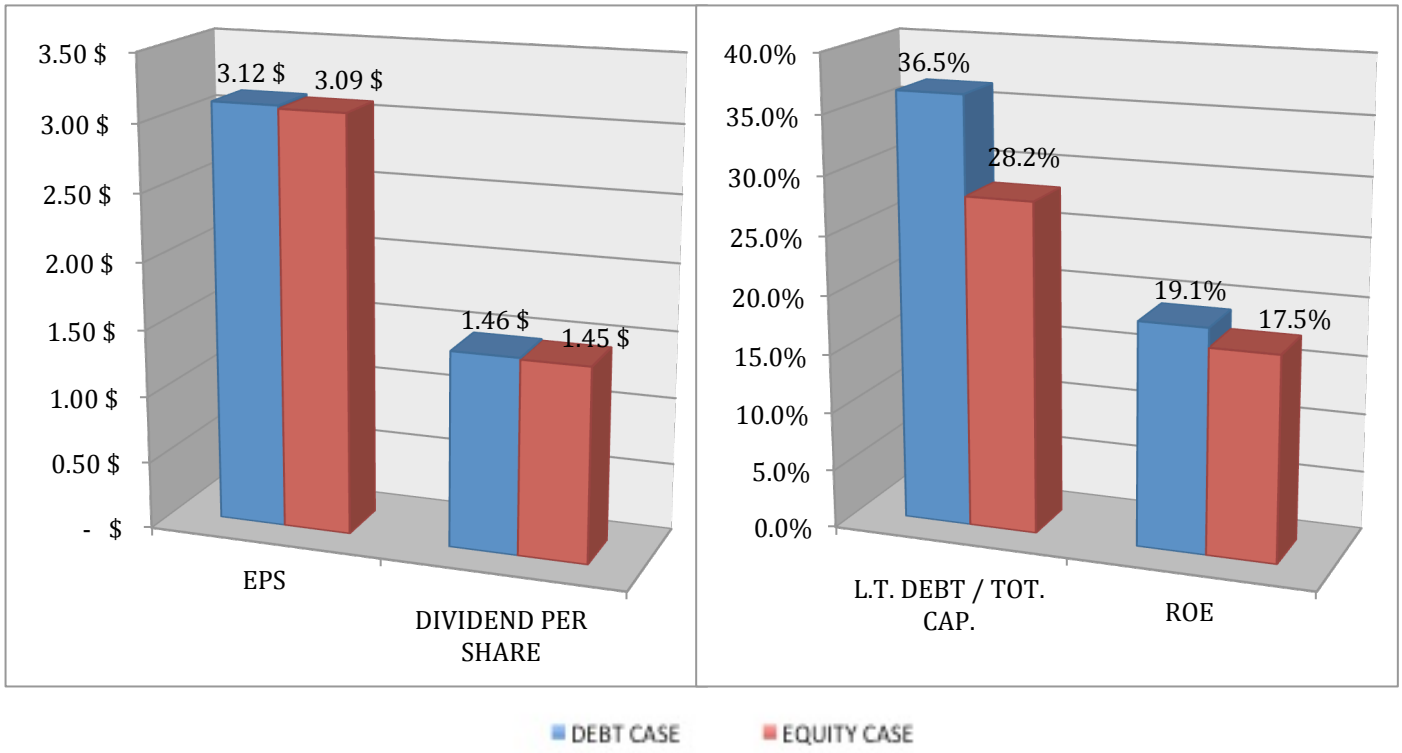


Exhibit 2a: 1995 Hutchison Whampoa's Capital Structure

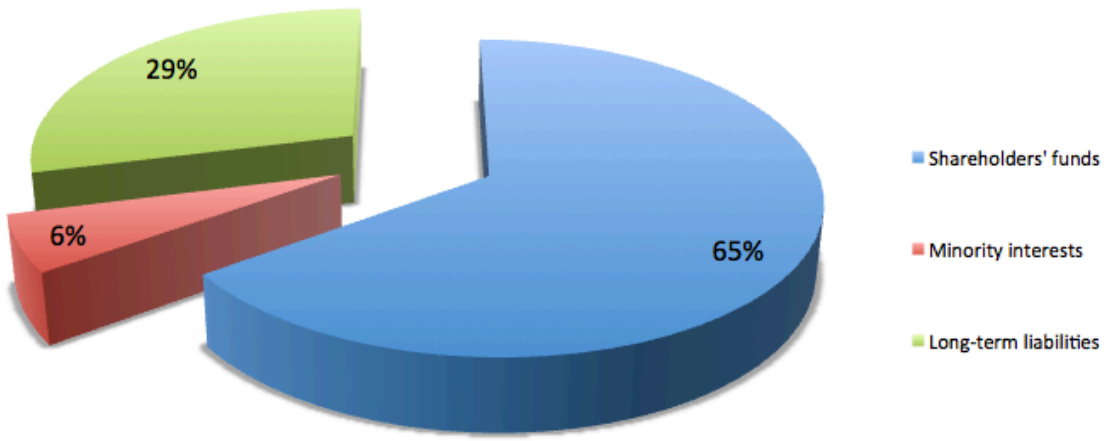


Exhibit 2b: Hutchison Whampoa's Forecasted Financial Position from 1996 to 2000

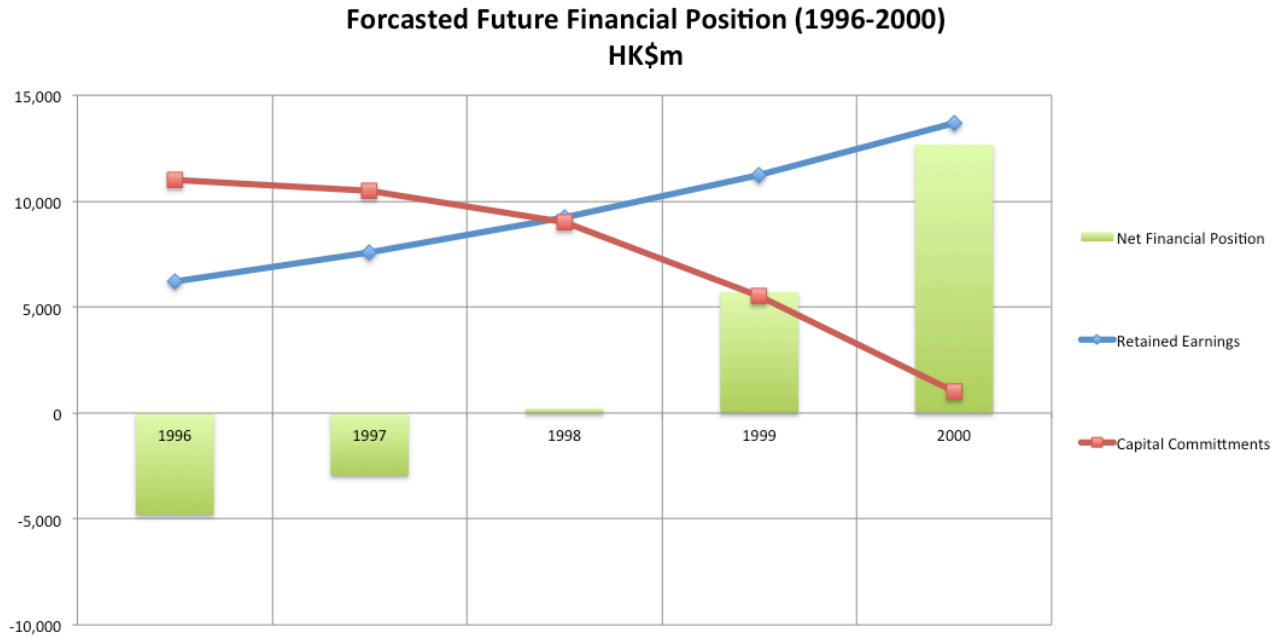


Exhibit 2c: Long Term Debt-to-Capital Ratio

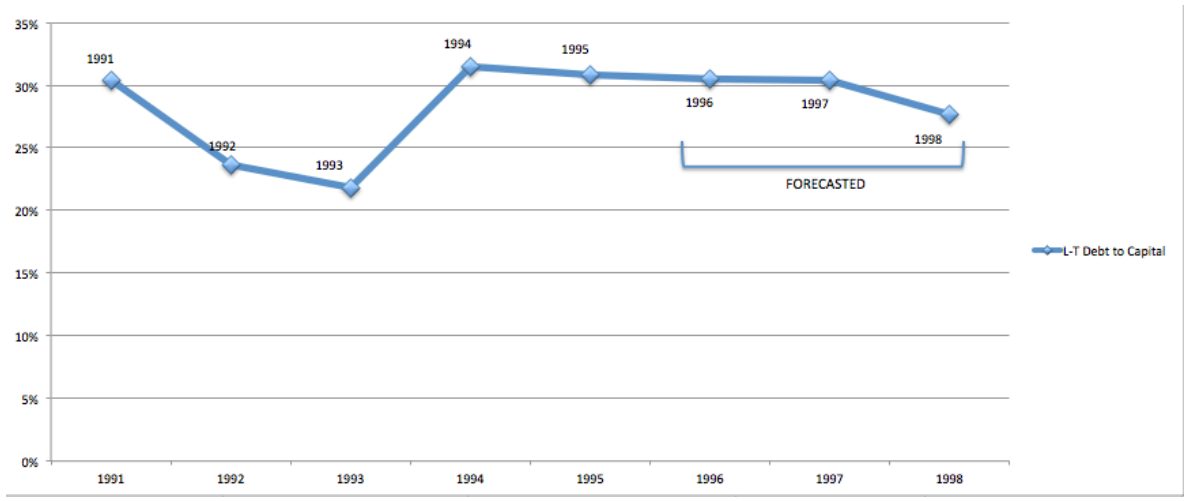


Exhibit 3: S&P's Hutchison Whampoa's BBB Rating.

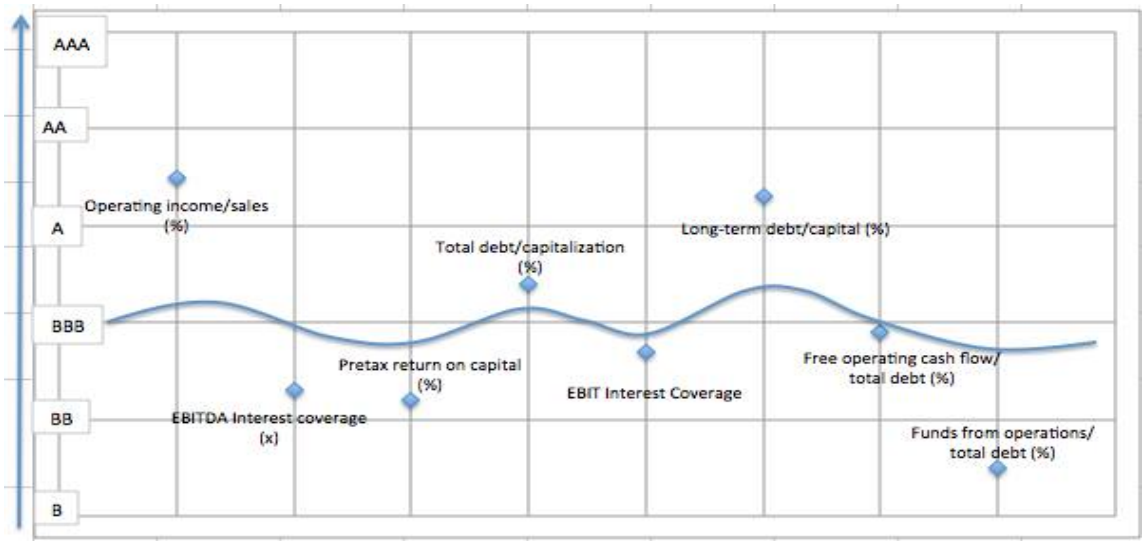
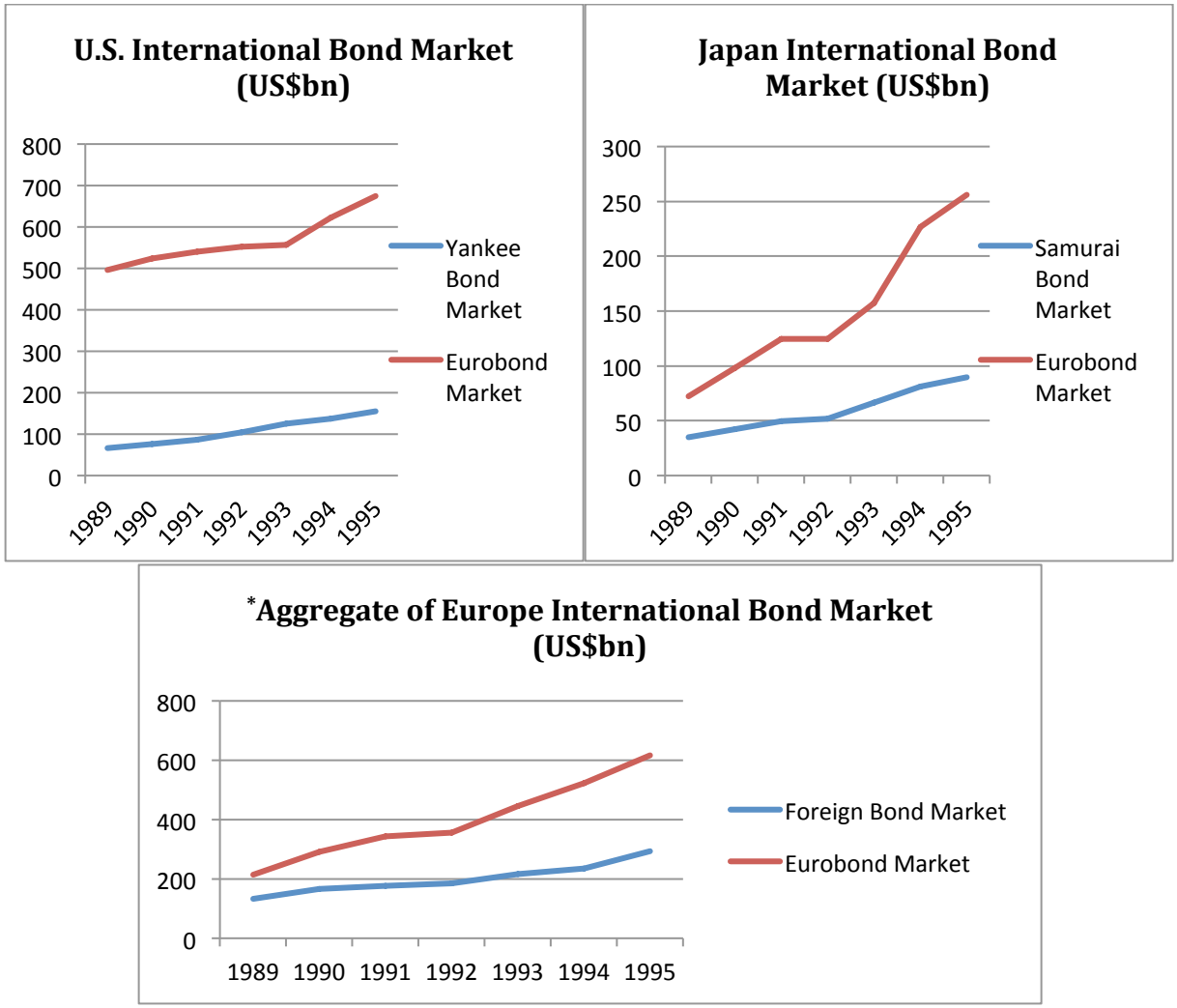


Exhibit 4a: Value Outstanding in Nominal Terms



*Aggregate of Europe= average of Germany, Italy, France and United Kingdom values.

Exhibit 4b: Market Growth Rate from 1989 to 1995

	International Bond Market	Foreign Bond Market	Eurobond Market
USA	+85.30%	+134.69%	+35.92%
Japan	+206.20%	**+158.33%**	+254.08%
*Aggregate of Europe	**+261.75%**	+36.44%	**+487.07%**
**World Aggregate	+184.42%	+109.82%	+259.02%

* Aggregate of Europe= average of Germany, Italy, France and United Kingdom growth rates.

**World Aggregate= average of USA, Japan and Aggregate of Europe growth rates

Executive Summary

Hutchison Whampoa Limited (the "Company") is one of the leading conglomerates in Asia with over HK$35,000 billion in revenues and with expanding global activities. The Company is planning to raise US$ 1 billion to sustain its growth rate in its diversified businesses.

This report analyses the different financing possibilities Hutchison Whampoa can resort to, to finance its growth in the near future. In our analysis we take into account the company´s capital structure and default risk. Methods of analysis include scenarios simulation, credit rating model and equity valuation multiple. Our report shows that HWL should issue Eurobonds denominated in German mark to fund their growth in 1996, because DEM is a stable currency and because of Germany's important role in Europe's economy.

These forecasts may be biased and will depend of the future of the global economy.

Summary of Facts

Hutchison Whampoa is a Hong Kong-based trading company. The company has a diversified portfolio and a market capitalization of US$ 26 billion. The firm has outstanding growth opportunities in the areas it operates in, with a US$ 2 billion property development project due to be finished by 2002.

Hong Kong firms relied in the past always more heavily on internally generated funds. (Analysts have determined that to maintain the current growth rate, Hutchison Whampoa would need five billion U.S dollars over the next five years). Hutchison Whampoa financed its past positive NPV projects from cash on hand, internal cash generation and short- and medium-term financing. The need for ten years funds and five billion U.S. dollars over the next five years cannot be realized with a syndicate of local and global banks to provide future capital needs on a floating interest rate.

Hutchison Whampoa Limited

Type	Public
Industry	Conglomerate
Founded	1863
Chairman	Li Ka-shing

Core Businesses:

- Property Development & Holding
- Ports & Related Services
- Retail & Manufacturing
- Telecommunication
- Energy & Infrastructure

Revenue	HK$ 35,026 million (1995)
Net income	HK$ 9,567 million (1995)
Owner	Cheung Kong HLD (49.9%)
Traded on	Hong Kong SE
Rating	BBB (expected)

Analysts, Class 17 & 18

Agatha Siomkos – *Greece*
Camilla Rettura – *Italy*
Djordje Mitić – *Austria*
Shubhi Jhalani – *India*
Tristan Le Gouz – *France*

WeWillBankYou GmbH *©2013*

Statement of Problem

After assessing the HWL's default risk, a choice has to be made between debt and equity. The company faces three questions:

- How would current main shareholder react to an issue of equity?
- Would the new capital structure be in adequation with the strong future growth of the Company?
- How should HWL raise the needed funds?

Analysis

The Company is facing a fundamental trade-off: to issue new equity or to issue long-term debt to raise funds. In order to solve this problem, we simulated future financial metrics (table 1 & exhibit 3). Our assumption of growth mirrors the last five years.

Looking at the earnings per share, issuing debt would be the best option available because of higher EPS and because the payout ratio stays constant. HWL is family held; the Li Ka-shing family receives half of the dividend payout each year. Issuing debt would please the owning family without hampering growth. Keeping the payout ratio stable is a strong signal to the investors that a solid dividend policy is maintained by the board of directors.

A seasoned equity offering, to finance the US$ 1 billion, would send a negative signal to the market, and dilute the participation of Li Ka-shing. By doing so, HWL would undertake a risk with the investors. The positive side of equity issuance would be that the share would be bought by local investors that know the company very well.

We used a model-driven approach to assess the long-term creditworthiness of Hutchison Whampoa Limited (table 2). We focus on selected financial ratios of comparable firms in Hong Kong. Anticipating Standard & Poor's rating, we believe that their opinion will be "Hutchison Whampoa´s obligations are rated 'BBB' and exhibit adequate protection parameters. However, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity of meeting its financial commitment on their obligations."

Several debt options (table 3) are available to HWL. Traditionally Hong Kong firms rarely issue debt to finance their activities, so the domestic market is underdeveloped to meet

the needs of HWL at the moment. An obstacle for the company is that it is not well known outside of Asia and particularly in the U.S.

Yankee bonds present a difficult scenario for the company. They are not convenient, because of the current regulations in the US. In order to issue them, the company would need to change its Chinese financial reports to US GAAP, which implies additional costs and a big loss in valuable time. Another problem that HWL should anticipate is that the BBB ranking would demand them to pay an 8.2% bond yield on the issued debt, which is from our perspective too high. On the other hand, US has the most developed bond market in the world, where raising this money would not be a problem. The market is bigger than all of the other 5 combined and the US dollar is the largest reserve currency and the most traded currency in the world. These arguments make the Yankee bond very appealing. HWL could use the second option to issue Yankee bonds, according to the SEC adopted rule 144A. This would not require a change to the financial statements but HWL is unknown to US institutional investors, and this option would not guarantee the financial needs to be met. HWL could also issue plain vanilla bonds in Hong Kong. This straight debt option has various advantages: it is easy to raise because the Company has a strong reputation, it requires no adjustments to financial statements and no additional costs. But the financing needs are too massive for the Hong Kong market, which is swamped by bond issues from PRC borrowers. This leads to an increasing upward pressure on pricing, which increases the cost of debt.

The option we would suggest for HWL is a Eurobond issue. The world market is a growing one, the par values are small and the market is highly liquid. There are no advertising costs involved because we aim to find institutional investors, looking to diversify their portfolios. The Company has to deal with foreign exchange risk but choosing a stable currency would reduce this risk. It will also face regular issue cost.

Recommendation

We would recommend for HWL to issue **Eurobonds** denominated in **German mark** to the international market (exhibit 5). German mark is a major reserve currency in the world and the dominating currency in Europe at the moment. It counts as a **very stable currency** and the European Exchange Rate Mechanism is based on the fixed exchange to the Deutsch mark.

The German bond market from 1989 to 1995 grew 2.7 times and is currently the third biggest bond market in the world, making it very attractive for investing in. The main benefits of the this choice are the **unregistered form**, that it can satisfy the **massive financial needs** of Hutchison Whampoa at a 6.5% bond yield and that the **foreign exchange risk** of DEM is lower because of the current European monetary policy of fixed exchange rates.

Exhibits and Tables

Table 1: Forward financial and valuation metrics

	FY1995	FY1996E w/equity	FY1996E w/debt
Net income	9 567 000 000 HKD	11 410 560 900 HKD	11 350 360 900 HKD
Debt	- HKD	- HKD	60 200 000 HKD
Shareholders' funds	58 839 000 000 HKD	59 839 000 000 HKD	58 839 000 000 HKD
Market cap	146 573 660 377 HKD	183 977 207 547 HKD	176 177 207 547 HKD
Market price	40,60 HKD	48,80 HKD	48,80 HKD
Shares outstanding	3 610 188 679	3 770 024 745	3 610 188 679
EPS	2,65	3,03	3,14
DPS	1,18	1,44	1,50
Dividends	4 267 000 000 HKD	5 415 676 400 HKD	5 415 676 400 HKD
Payout ratio	44,53%	47,46%	47,71%
ROE	16,26%	19,07%	19,29%
P/E	15,3	16,1	15,5
Long-term debt	26 174 000 000 HKD	26 174 000 000 HKD	26 234 200 000 HKD
Long-term debt/total capital	0,31	0,30	0,31

Source: Hutchison Whampoa Limited Annual Report, DataStream, WWBY Research

Table 2: Credit rating model

Financial ratios	HWL	S&P	Swire	S&P	Wharf	S&P
EBITDA interest coverage	6,2	BBB	5,5	BBB	4,5	BB
EBIT interest coverage	4,1	BBB	3,7	BBB	3,2	BBB
Funds from operations/total debt	32,8	BBB	29,4	BBB	15,2	BB
Free operating cash flow/total debt	9,4	BBB	17,5	A	61,7	AAA
Pretax return on capital	12,0	BB	8,1	B	3,9	B
Operating income/sales	16,4	A	12,4	B	n/a	n/a
Long-term debt/capital	29,2	A	26,5	A	21,5	A
Total debt/capitalization	40,8	A	31,9	AA	23,1	AAA
Long-term debt ratings		BBB		A		A

Note: Median of FY1994-95 for every ratio. Weighted average to determine overall rating with B weighted 1 and AAA weighed 6.
Source: Compustat, Bloomberg, Standard and Poor, 1999

Table 3: Available debt options

Criterions	Straight Bonds	Eurobonds	Yankee bonds 'SEC 1933 '	Yankee Bonds '144A'
Institutional investors		X		X
Unregistered form		X		X
Periodic reporting			X	
Reconciled to U.S. GAAP			X	
Massive financing needs		X	X	X
Foreign Exchange risk		X		
Choice		X		

Source: WeWillBankYou Research

Exhibit 4: Growth (EPS)

Source: HWL Annual Report, WWBY Research

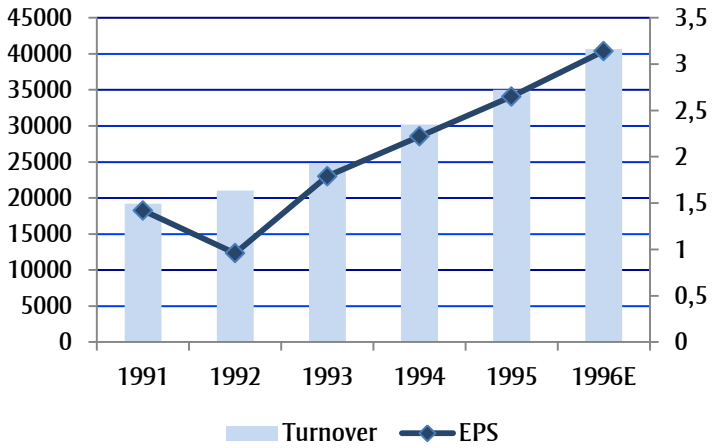
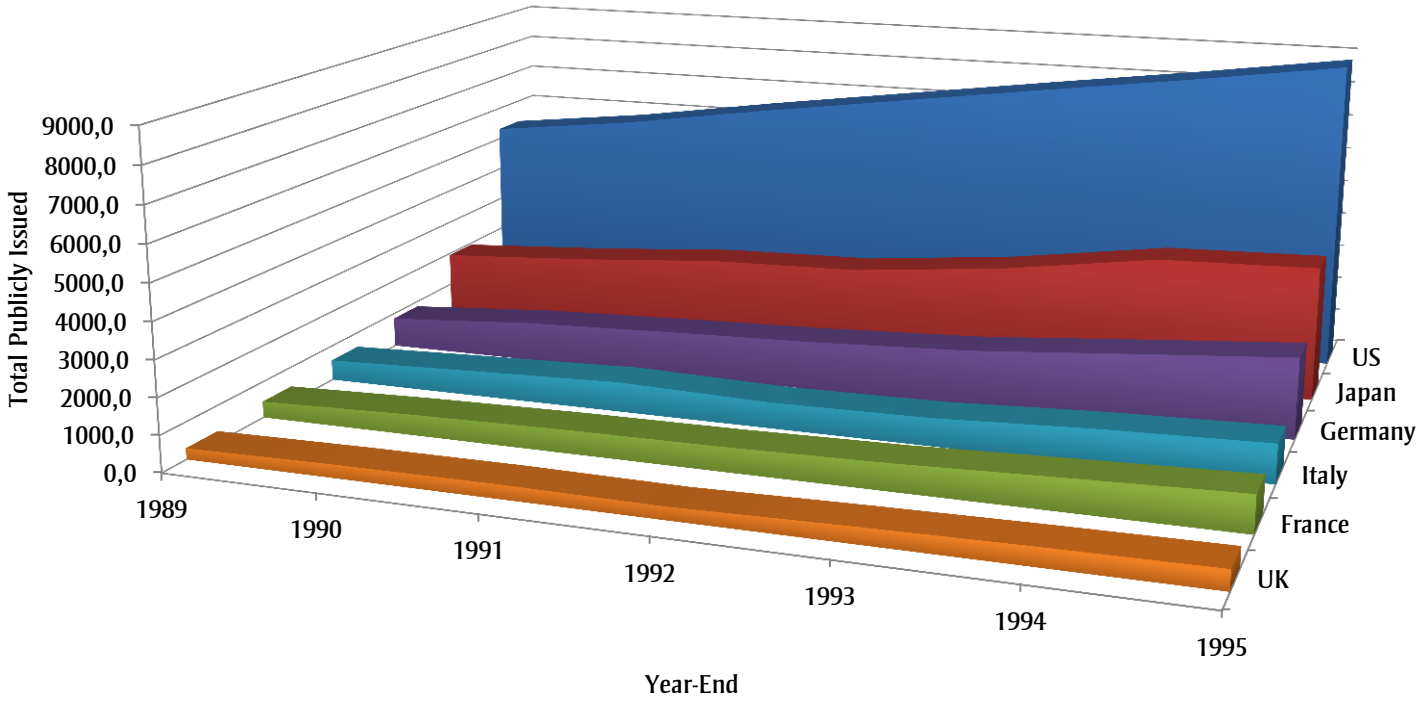


Exhibit 5: Growth of the six largest bond markets 1989 to 1995

Source: "How Big is the World Bond Market –1996 Update." Salomon Brothers Inc.



ANALYSTS CERTIFICATIONS

We, Agatha, Camilla, Shubhi, Djordje and Tristan, hereby certify that the views expressed in this case study report accurately reflect our personal views about any or all of the subject securities or issuers referred to in this case study report and no part of our compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this case study report.